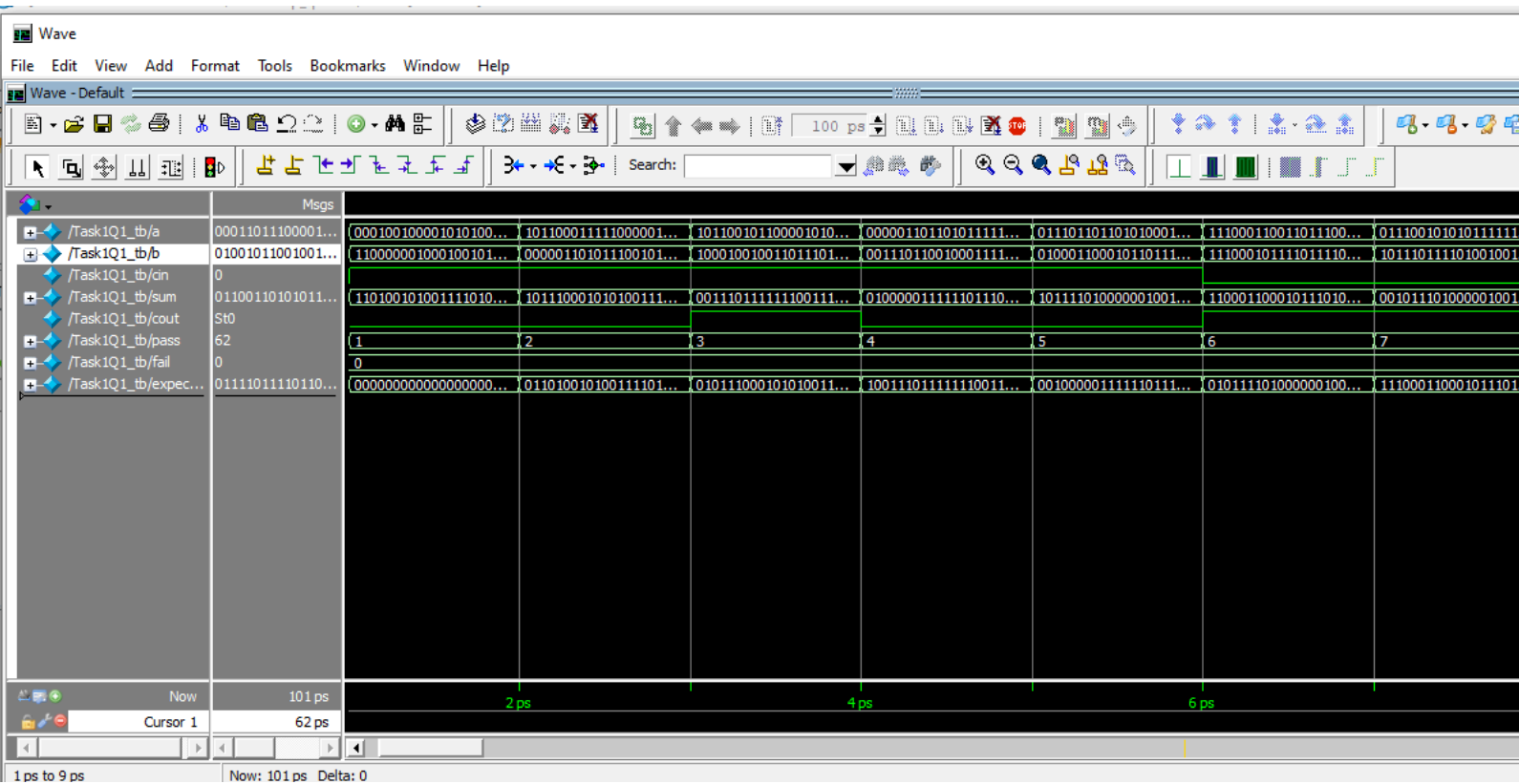


Q1 adder32b Simulation Waveforms



Type	ID	Message

		Running Quartus Prime Analysis & Elaboration
		Command: quartus_map --read_settings_files=on --write_settings_files=off Task1Q1 -c Task1Q1 --analysis_and_elaboration
		18236 Number of processors has not been specified which may cause overloading on shared machines. Set the global assignment NUM_PARALLEL
		20030 Parallel compilation is enabled and will use 2 of the 2 processors detected
		12021 Found 1 design units, including 1 entities, in source file task1q1.v
		12021 Found 1 design units, including 1 entities, in source file task1q1_tb.v
		12127 Elaborating entity "Task1Q1" for the top level hierarchy
		Quartus Prime Analysis & Elaboration was successful. 0 errors, 1 warning

Quartus Prime Lite Edition - D:/internship_quartus/Task1Q1 - Task1Q1

File Edit View Project Assignments Processing Tools Window Help

add.v Task1Q1 Compilation Report - Task1Q1 Task1Q1.v Task1Q1_tb.v Task1Q2.v

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Flow Summary

Flow Status: Successful - Sun Dec 21 20:25:57 2025

Quartus Prime Version: 18.1.0 Build 625 09/12/2018 SJ Lite Edition

Revision Name: Task1Q1

Top-level Entity Name: Task1Q1

Family: Cyclone V

Device: 5CEFA7F3117

Timing Models: Final

Logic utilization (in ALMs): N/A until Partition Merge

Total registers: N/A until Partition Merge

Total pins: N/A until Partition Merge

Total virtual pins: N/A until Partition Merge

Total block memory bits: N/A until Partition Merge

Total PLLs: N/A until Partition Merge

Total DLLs: N/A until Partition Merge

Messages

System (6) Processing (26)

Activated Go to Se

Q2 Barrel Shifter results

Quartus Prime Lite Edition - D:/Task1Q2/Task1Q2 - Task1Q2

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Task1Q2

Compilation Report - Task1Q2 Task1Q2.v Task1Q2_tb.v

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Flow Summary

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Flow Status	Successful - Sun Dec 21 21:10:48 2025
Quartus Prime Version	18.1.0 Build 625 09/12/2018 SJ Lite Edition
Revision Name	Task1Q2
Top-level Entity Name	Task1Q2
Family	Cyclone V
Device	5CEFA7F3117
Timing Models	Final
Logic utilization (in ALMs)	250 / 56,480 (< 1 %)
Total registers	0
Total pins	70 / 480 (15 %)
Total virtual pins	0
Total block memory bits	0 / 7,024,640 (0 %)
Total DSP Blocks	0 / 156 (0 %)
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	

	Msgs	
/Task1Q2_tb/data_in	xxxxxxxxxxxxxxxx...	00010010001101000101011001111000
/Task1Q2_tb/shift_amt	xxxxxx	01000
/Task1Q2_tb/dir	x	
/Task1Q2_tb/data_out	xxxxxxxxxxxxxxxx...	00110100010101100111100000010010
/Task1Q2_tb/pass	0	0
/Task1Q2_tb/fail	0	4
/Task1Q2_tb/expected	xxxxxxxxxxxxxxxx...	00110100010101100111100000010010

	Msgs	
/Task1Q2_tb/data_in	xxxxxxxxxxxxxxxx...	000000000000000000000000000000001111
/Task1Q2_tb/shift_amt	xxxxxx	00000
/Task1Q2_tb/dir	x	
/Task1Q2_tb/data_out	xxxxxxxxxxxxxxxx...	000000000000000000000000000000001111
/Task1Q2_tb/pass	0	0
/Task1Q2_tb/fail	0	1
/Task1Q2_tb/expected	xxxxxxxxxxxxxxxx...	000000000000000000000000000000001111

	Msgs	
/Task1Q2_tb/data_in	xxxxxxxxxxxxxxxx...	000000000000000000000000000000001
/Task1Q2_tb/shift_amt	xxxxxx	11111
/Task1Q2_tb/dir	x	
/Task1Q2_tb/data_out	xxxxxxxxxxxxxxxx...	100000000000000000000000000000000
/Task1Q2_tb/pass	0	0
/Task1Q2_tb/fail	0	5
/Task1Q2_tb/expected	xxxxxxxxxxxxxxxx...	100000000000000000000000000000000

Q3 Priority Encoder

Task1Q3

Task1Q3.v

Task1Q3_tb.v

Compilation Report - Task1Q3

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Flow Status

Successful - Sun Dec 21 22:15:28 2025

Quartus Prime Version

18.1.0 Build 625 09/12/2018 SJ Lite Edition

Revision Name

Task1Q3

Top-level Entity Name

Task1Q3

Family

Cyclone V

Device

5CEFA2F23C6

Timing Models

Final

Logic utilization (in ALMs)

4 / 9,430 (< 1 %)

Total registers

0

Total pins

12 / 224 (5 %)

Total virtual pins

0

Total block memory bits

0 / 1,802,240 (0 %)

Total DSP Blocks

0 / 25 (0 %)

Total HSSI RX PCSs

0

	Msgs	
/Task1Q3_tb/in	00001100	10000000
/Task1Q3_tb/out	011	111
/Task1Q3_tb/valid	St1	
/Task1Q3_tb/expec...	011	111
/Task1Q3_tb/expec...	1	

	Msgs	
/Task1Q3_tb/in	00001100	00001100
/Task1Q3_tb/out	011	011
/Task1Q3_tb/valid	St1	
/Task1Q3_tb/expec...	011	011
/Task1Q3_tb/expec...	1	